

	JERUSALEM COLLEGE OF ENGINEERING (An Autonomous Institution, Affiliated to Anna University, Chennai)
	QUESTION BANK

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Subject Name: Environmental Science & Sustainability

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III	Renewable sources of Energy			
IV	Sustainable Development			
V	Sustainability Practices			

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Unit 1 - ENVIRONMENT AND BIODIVERSITY

Cob1 – To study the inter-relationship between living organisms and environment.

CO1 – To recognize the function of ecosystem and contribute to conservation of biodiversity.

Bloom's Taxonomy Level: K1, K2, K3, K4, K5

Part A

1	Mention the scope & significance of Environmental studies.	CO1	K2
2	Define ecosystem. What are its types?	CO1	K1
3	Energy flow is always unidirectional. Why?	CO1	K3
4	What is meant by ecological pyramid? What are its types?	CO1	K2
5	Draw the structure of ecological pyramid of grassland ecosystem.	CO1	K4
6	Draw the structure of ecological pyramid of parasite ecosystem.	CO1	K4
7	Define biodiversity. What are its types?	CO1	K2
8	What are the two main functions of biodiversity?	CO1	K3
9	What are biodiversity hotspots?	CO1	K4
10	What are endemic species and endangered species?	CO1	K2
11	What is RED data book?	CO1	K3
12	What are endemic and endangered species?	CO1	K3

Part B

1	Briefly explain the structural and functional components of an ecosystem.	CO1	K3
2	Briefly explain the energy flow through ecosystem.	CO1	K4
3	Discuss the concept of ecological succession.	CO1	K5
4	Discuss ecological pyramid with suitable example.	CO1	K3
5	Name and briefly describe two hot spots of biodiversity in India	CO1	K4
6	Explain the various threats to Biodiversity.	CO1	K3
7	Explain values of biodiversity.	CO1	K4
8	Explain In-situ and Ex- situ conservation of biodiversity.	CO1	K4
9	Write a note on: India as Mega diversity	CO1	K3

Unit 2 – ENVIRONMENTAL POLLUTION

Cob2 – To learn causes, effects and preventive methods of environmental pollution.

CO2 – To identify the causes and effects of environmental pollution and practice the preventive measures.

Bloom's Taxonomy Level: K2, K3, K4, K5

Part A

1	What are the different sources of air pollution?	CO2	K3
2	What are point & non-point sources of water pollution?	CO2	K2
3	Define soil pollution. Write any two causes of soil pollution.	CO2	K1, K2
4	Define Thermal pollution. What are the effects of thermal pollution?	CO2	K2
5	What is Eutrophication?	CO2	K3
6	What is blue baby syndrome?	CO2	K3
7	What are the objectives of wild life protection Act?	CO2	K3
8	List out the objective of forest conservation Act.	CO2	K3
9	List various clean air Act amendments.	CO2	K4
10	What are e-wastes?	CO2	K2
11	Expand OHSMS and define it.	CO2	K2
12	What is the scope of the Ohsms?	CO2	K2

PART-B

1	Discuss the major air pollutants and their impacts on environment.	CO2	K4
2	Explain the causes, effects and control measures of water pollution.	CO2	K5
3	What is Thermal pollution and explain their impacts?	CO2	K4
4	Describe the causes, effects and control measures of water pollution.	CO2	K5
5	What are the salient features of environmental (protection) Act 1986.	CO2	K5
6	Discuss the methods of disposal of municipal solid waste management.	CO2	K4
7	Explain how and why electronic waste (e-waste) has become a growing solid waste problem.	CO2	K4
8	Write a note on OHSMS with a case study.	CO2	K3
9	Explain a case study on water pollution and air pollution.	CO2	K5

Unit 3 - RENEWABLE SOURCES OF ENERGY

Cob3 – To impart knowledge on the needs of renewable and new energy sources.

CO3 – To recognize various types of new renewable sources of energy and their potential applications.

Bloom's Taxonomy Level: K1, K2, K3, K4, K5

PART A

1	What is the principle of energy management?	CO3	K3
2	What are the objectives of energy management?	CO3	K3
3	What are the objectives of energy conservation?	CO3	K3
4	Why we need new energy sources?	CO3	K2
5	What are the advantages of renewable energy?	CO3	K3
6	How are energy resources classified?	CO3	K1
7	What are the limitations of using hydrogen as a new energy source?	CO3	K3
8	What are the sources of hydrogen energy?	CO3	K2
9	Comment on the important applications of hydrogen energy.	CO3	K4
10	Highlight about the possibilities of the new energy sources suitable for the implementation.	CO3	K5
11	List out the advantages and disadvantages of OTE, GTE and Tidal E.	CO3	K3

PART B

1	Summarize the objectives, principle, importance and the ways through energy conservation is made.	CO3	K3
2	Explain the principle and various steps involved in the energy management.	CO3	K3
3	Write a note on (i) tidal energy (ii) OTE (iii) GTE	CO3	K3
4	Hydrogen, a new energy source- Justify.	CO3	K5

Unit 4 - SUSTAINABLE DEVELOPMENT

Cob4– To acquire broad knowledge on sustainable development and its values on society.

CO4 – To apply the sustainable development goals for technological advancement and societal development.

Bloom's Taxonomy Level: K1, K2,

PART A

1	What is sustainable development?	CO4	K2
2	Why we need a sustainable development?	CO4	K2
3	What are the challenges to overcome the sustainable development?	CO4	K3
4	List out any 5 millennium development goals for achieving the sustainability.	CO4	K3
5	Differentiate sustainability and unsustainability.	CO4	K3
6	Define carbon credit.	CO4	K1
7	What is carbon footprint?	CO4	K2
8	Name any two protocols for sustainable development.	CO4	K3
9	What are the targets for sustainability?	CO4	K3
10	What are the indicators for developing sustainable development?	CO4	K5

PART B

1	Write about economic and social challenges of sustainability.	CO4	K5
2	List out the goals of sustainable development	CO4	K3
3	What are the indicators for sustainable development?	CO4	K3
4	Mention the intervention area of sustainable development	CO4	K3
5	Write a brief notes on (i) climate change in Chennai (Flood)	CO4	K5
6	List out the possible solutions for climate change.	CO4	K5
7	How to lower the carbon footprint?	CO4	K5
8	What are the advantages of carbon footprint and carbon credit?	CO4	K3

Unit 5 - SUSTAINABILITY PRACTICES

Cob5 – To expertise sustainability practices of sustainable habitat, sustainable energy and green engineering.

CO5 – To adopt sustainability practices for energy efficiency and sustainable urbanization.

Bloom's Taxonomy Level: K1, K2, K3, K4, K5

PART A

1	Define zero waste.	CO5	K2
2	What are the steps to achieve zero waste?	CO5	K5
3	Explain the concepts of 5R.	CO5	K3
4	Define ISO 14000 series.	CO5	K2
5	What are the advantages and disadvantages of ISO 14000 series?	CO5	K2
6	Define Life cycle Assessment.	CO5	K2
7	What is EIA? What are its objectives?	CO5	K1
8	What are the key elements of EIA?	CO5	K2
9	What are the characteristics of sustainable habitat?	CO5	K2
10	How to promote sustainable transport?	CO5	K4
11	How we can calculate energy efficiency for achieving sustainability?	CO5	K5

PART B

1	What are the necessary steps to achieve the circular economy?	CO5	K5
2	Give brief notes on green building.	CO5	K4
3	Discuss the benefits and limitation of sustainable transport.	CO5	K3
4	Write a note of carbon emission and sequestration.	CO5	K4
5	What are the principle followed for green engineering?	CO5	K2
6	What are the technological change on sustainable urbanization?	CO5	K2
7	Explain the rules to develop the sustainable urbanization?	CO5	K4